Git Commands

Staging & Committing

```
git status                          # See modified, untracked, and staged files
git add <file>                      # Stage a specific file for the next commit
git add                             # Stage all modified and new files
git commit -m "Commit message"      # Commit staged changes with a descriptive message
git commit --amend                  # Modify the last commit message or add forgotten files
```

Branching & Merging

```
git branch                          # List local branches (* indicates current branch)
git branch -a                       # List both local and remote tracking branches
git branch <branch-name>            # Create a new branch
git switch <branch-name>            # Switch to an existing branch (recommended over checkout)
git switch -c <branch-name>         # Create a new branch and switch to it immediately
git merge <branch-name>             # Merge the specified branch into your active branch
git branch -d <branch-name>         # Delete a branch safely (if fully merged)
git branch -D <branch-name>         # Force delete a branch (even if unmerged)
```

Git Commands

Sharing & Syncing with Remotes

```
git remote add origin <repo-url>      # Map a remote repository URL to the shortcut "origin“
git remote -v                         # View configured remote connections
git fetch <remote>                    # Pull down history from remote without merging changes
git pull                              # Fetch changes from remote and immediately merge them
git push -u origin <branch-name>      # Push commits to remote - remember the tracking
git push                              # Push local commits to the tracked remote branch
```

Inspecting History & Diffs

```
git log                               # Show complete commit history
git log --oneline                     # View commit history condensed to single lines
git log --graph --oneline --all       # Visualize branch structures as an ASCII graph text
git diff                              # Show unstaged modifications vs. your current index
git diff --staged                     # Show staged modifications ready for a commit
git blame <file>                      # Line-by-line view tracking who made changes and when
```

Fundamental Git Commands

Purposes	Commands
To check the version	git --version
Configure your Git username	git config --global user.name "Your Name"
Configure your email	git config --global user.email "your-email@example.com"
Check your configuration	git config --global --list
Create a folder for your project	mkdir foldername
Move into the folder	cd foldername
Check the current directory	pwd /cd
Initialize a Git repository	git init
Check the status of your project	git status
To stage a particular file	git add main.py
To stage all changes in the current directory	git add .

Fundamental Git Commands

Purposes	Commands
Commit your changes	git commit -m "Initial commit"
View commit history	git log
For a shorter commit history	git log --oneline
View changes in files	git diff
To view staged changes	git diff --staged
To stop tracking a file and remove it from the working directory	git rm main.py
Then commit the deletion	git commit -m "Removed main.py"
Rename a file	git mv old_name.py new_name.py
Connect your local repository to GitHub	git remote add origin https://github.com/YOUR-USERNAME/foldername.git
Check the remote repository	git remote -v

Fundamental Git Commands

Purposes	Commands
Upload your project to GitHub:	
First, ensure your changes are committed	git add . git commit -m "Initial project version"
Set your primary branch to main	git branch -M main
Upload your project	git push -u origin main
Download an existing GitHub project:	
To download a local copy	git clone https://github.com/OWNER/REPOSITORY.git
Download the latest changes	git pull
To specify the remote and branch	git pull origin main
Upload your latest changes	git add . git commit -m "Updated Content" git push

Fundamental Git Commands

Purposes	Commands
To List branches	git branch
Create a new branch	git branch branch-name
Switch to a branch	git switch branch-name
Create and switch to a new branch	git switch -c branch-name
Switch back to the main branch	git switch main
Merge a branch: First, switch to the branch into which you want to merge the changes Then merge the feature branch	 git switch main git merge branch-name

Fundamental Git Commands

Purposes	Commands
View remote repository addresses	git remote -v
Clone a remote repository	git clone URL
Fetch and integrate remote changes	git pull
Upload local commits	git push
Download remote updates without integrating them	git fetch
Discard unstaged changes to a tracked file	git restore filename
Remove a file and stage its removal	git rm filename
Rename or move a tracked file	git mv old new